

# AI-Assisted Systematic Reviews and Meta-Analysis

## Day 3: From Studies to Evidence

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instats Seminar

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# Day 3: From Studies to Evidence

- 1 The Challenge of Data Extraction
- 2 Designing the Extraction Schema
- 3 Synthesis: Narrative vs. Quantitative
- 4 Reporting Standards
- 5 Practical Session and BYOD

# From Text to Structured Data

By the end of Day 2, we have a definitive list of included studies. The final and most critical phase of the systematic review is extracting the actual findings from those studies to synthesize the evidence.

Manual data extraction requires researchers to read full-text PDFs and manually transcribe sample sizes, methodologies, PICO elements, and effect sizes into spreadsheets. This is highly prone to transcription errors and extremely time-consuming.

# LLMs for Structured Extraction

Large Language Models excel at parsing unstructured text and mapping it to a predefined schema.

Recent studies (Mahmoudi et al., 2025) demonstrate that LLMs can achieve high accuracy on structured extraction tasks. By prompting the model to output a strict JSON object containing specific fields, we can programmatically convert abstracts (and eventually full texts) into clean tabular data.

However, the persistent risk of **hallucination** — particularly inventing numbers or effect sizes that are not in the text — means that human spot-checking remains essential.

# Discipline-Specific Schemas

An extraction schema is a list of variables you want to collect from every study. The schema must be tailored to the research question and the discipline.

## Health Sciences (PICO):

- Population (e.g., sample size, demographics)
- Intervention (e.g., drug dosage, telemedicine)
- Comparator (e.g., placebo, standard care)
- Outcome (e.g., Risk Ratio, Odds Ratio, 95% CI)

## Management / Business:

- CSR Measure (e.g., ESG score)
- Financial Performance Metric (e.g., ROA, Tobin's Q)
- Correlation Coefficient ( $r$ )

# Prompting for JSON Output

To ensure the LLM returns data we can automatically convert into a CSV, we use a highly structured prompt:

You are a systematic review data extractor. Extract the following fields from the study below.

Return your answer as a valid JSON object with exactly these keys:

- Population: string
- Sample\_Size: integer
- Effect\_Size: float

If a field cannot be determined, use null.

This forces the model into a deterministic formatting mode, dramatically reducing conversational hallucinations.

# Narrative Synthesis

Not all systematic reviews lead to a meta-analysis. If the included studies use widely different methodologies, populations, or outcome measures, pooling them statistically is inappropriate.

In these cases, we perform a **Narrative Synthesis**:

- Organizing the extracted data into a structured summary table
- Grouping studies by theme, intervention type, or methodology
- Providing a plain-language summary of the direction of effects

(Example: Our Social Sciences case study on Universal Basic Income).

# Quantitative Synthesis (Meta-Analysis)

When studies report comparable quantitative effect sizes (e.g., Risk Ratios, Odds Ratios, or Correlation Coefficients) along with measures of variance (e.g., Confidence Intervals or Standard Errors), we can perform a **Meta-Analysis**.

The core computation is the **Inverse-Variance Weighted Mean**:

- Studies with larger sample sizes (tighter confidence intervals) are given more weight.
- Studies with smaller sample sizes (wider confidence intervals) are given less weight.
- The result is a single "Pooled Estimate" of the true effect.

# The Forest Plot

The standard visualization for a meta-analysis is the **Forest Plot**.

- Each horizontal line represents a single study.
- The square in the middle is the study's effect size.
- The width of the line is the 95% Confidence Interval.
- The diamond at the bottom is the **Pooled Estimate**.
- The vertical line at 1.0 (for ratios) or 0.0 (for differences) is the "line of no effect."

Our workflow automatically generates Forest Plots for the Health Sciences and Management case studies using Python (`matplotlib`) and R (`metafor`).

# The PRISMA 2020 Flow Diagram

Ethical reporting of a systematic review requires documenting the attrition of records at every stage of the pipeline. The global standard is the **PRISMA 2020 Flow Diagram**.

The diagram tracks:

- 1 Records identified via databases/APIs
- 2 Records removed before screening (duplicates)
- 3 Records screened (title/abstract)
- 4 Records excluded
- 5 Full-text articles assessed for eligibility
- 6 Studies included in the final synthesis

Our Streamlit app automatically generates this diagram based on the record counts from your pipeline.

# Concept Network of Included Studies

Once extraction is complete, the app generates an **interactive concept network** (Step 5) from the structured extraction table.

- **Nodes** represent key concepts extracted from the Title, Country, Population, Intervention, and Outcome columns.
- **Edges** connect concepts that co-occur within the same study.
- Node size and colour encode concept frequency across the included studies.
- The network reveals the **conceptual architecture** of the evidence base — which themes cluster together and which are peripheral.

This complements the Forest Plot and PRISMA diagram by providing a **qualitative map** of the synthesised evidence, useful for identifying research gaps and framing the discussion section.

# Hands-On: Extraction and Synthesis

In the final practical session, we move to the Streamlit app.

**Guided Examples:** We will review the pre-extracted structured data for our four case studies. We will generate narrative summary tables, compute pooled effect sizes, draw Forest Plots, generate PRISMA diagrams, and Concept Network of included studies.

**Bring Your Own Data (BYOD):** You will upload your "Included Studies" CSV from Day 2. You will define your own extraction schema, prompt the LLM to extract the data, and generate your own synthesis outputs (Narrative Table, Forest Plot, PRISMA Diagram, and Concept Network of included studies).

# References

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# Questions and Discussion

Thank you!

Questions?

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